MapleCFO: Stage 2 Implementation Report

EECS 3311 Software Design, Fall 2026 · Krishna Patel
Repository: https://github.com/Krishna7997/EECS-3311-Project · Videos: see Section 7

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1. Project overview and implementation status

What it is. MapleCFO is a personal-finance agent for Canadian students and young professionals. Users import the CSV files their bank lets them download (no bank passwords, no bank connection). MapleCFO categorizes the transactions and answers practical questions: am I on budget, which subscriptions went up, how much TFSA/RRSP/FHSA room do I have, when could I retire early, which debt should I pay first, can I afford this, and where should extra money go.

Users. One person on their own computer. Data stays in a local SQLite database (`~/.maplecfo/maplecfo.db`). Only small snippets are sent to the AI model: unknown merchant names, the first lines of an unknown CSV, chat questions with their tool results, and summary figures.

Interfaces. A JavaFX GUI (13 pages, AtlantaFX theme) and a picocli CLI (one-shot commands and an interactive mode). Both use the same `MapleCfoFacade`, so every feature works in both.

AI / LLM integration. Anthropic Claude through the `LlmClient` interface (`ClaudeClientAdapter`; model `claude-haiku-5-5` by default, set in `.env`). The AI is used as an **agent**, not a text box:

- **F10 AI CFO chat:** a bounded reason-act loop (`CfoAgent`, up to 6 steps). The model chooses among 12 finance tools, reads their JSON results and answers. Every dollar amount in the answer is checked against the tool results (`ResponseValidator`); changes are only *proposed* and wait for the user's confirmation (Command pattern, undoable).
- **F17 AI monthly review:** a longer multi-step task (up to 10 steps) where the agent plans its own steps, gathers the figures, proposes up to five budgets and writes a to-do list.
- **F02, F12, F16, F18, F19:** the AI categorizes unknown merchants, writes the monthly summary, reads unknown bank formats, turns plain-English questions into filters and explains the extra-money plan. In each case Java checks the AI's answer before using it.

Without an API key the app runs in a clearly labelled **demo mode** (`MockLlmClient` with keyword rules), so it can be installed and tried offline. Demo mode is a fallback, not the AI feature: the videos use the real model.

Status. All 15 Stage 1 features are implemented and working, and four new AI features were added (F16-F19), for 19 features in total. 280 JUnit tests pass. 8 design patterns from the course are implemented (Section 3). Limitations are listed honestly in Section 6.4.

	Count
Features (Stage 1 + new)	15 + 4 = 19
Java source files (main)	200 in 21 packages
JUnit tests	280 (all passing), 32 test classes
Line coverage (JaCoCo)	59.7% overall, 89.7% excluding the JavaFX GUI package
Code commits Oct 20-26 / Oct 27-Nov 2 / Nov 3-9	5 / 7 / 28

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2. Feature to implementation mapping

Type: **D** deterministic, **AI** AI-based, **H** hybrid. Video codes refer to Section 7 (A = GUI video, B = CLI video).

ID and name	Stage 1	Status	Implemented classes / files and responsibilities	Key methods	Patterns	Type	Video
F01 Import bank CSVs	F01	Modified (format detection added)	`TransactionService` (import, duplicates, events); `CsvTransactionAdapter` (reads a CSV with a `ColumnMapping`); `DefaultProfiles` (TD, RBC, generic); `ImportResult`	`importFrom`, `detectProfile`, `importFile`, `matchesFirstRow`	Adapter, Observer	D	A3, B1
F02 Categorize and correct	F02	Implemented	`CategorizationService` (ordered chain); `RuleBasedCategorizer` + `DefaultRules` (~120 Canadian merchants, learned rules); `LlmCategorizer` (AI fallback); `RecategorizeCommand`	`categorizeAll`, `categorize`, `learn`, `recategorize`	Strategy, Command	H	A5, B3
F03 Accounts and net worth	F03	Implemented	`Account`, `AccountGroup`, `AssetComponent`; `AccountService`; `NetWorthService` (snapshots)	`buildAccountTree`, `getValue`, `computeSnapshot`	Composite	D	A9, B7
F04 Budgets and alerts	F04	Implemented	`BudgetService` (listens to imports, 80%/100% alerts); `SetBudgetCommand`; `AlertService`	`setBudget`, `getStatus`, `onEvent`	Observer, Command	D	A7, B5
F05 Recurring charges	F05	Modified (price steps, cancelled charges)	`RecurringChargeDetector`; `RecurringCharge`	`detect(txs, today)`, `build`	-	D	A8, B6
F06 TFSA/RRSP/FHSA room	F06	Modified (HBP/LLP, FHSA transfer, RRSP allowance)	`RegisteredAccountService`; `ContributionLimits`; `ContributionRoom`; `ContributionCheck`; `RepaymentPlan`; `ContributionKind`	`computeRoom`, `previewContribution`, `recordContribution`, `repaymentPlans`	-	D	A13, B9
F07 FIRE projection	F07	Modified (own target, target age, Coast FIRE)	`FireCalculator`; `FireInputs`; `FireProjection`	`project`, `yearsToFire`, `savingNeeded`	-	D	A16, B10
F08 Debt payoff	F08	Implemented	`DebtPayoffPlanner`; `PayoffStrategy`, `AvalancheStrategy`, `SnowballStrategy`; `PayoffPlan`	`plan`, `orderDebts`	Strategy	D	A15, B10
F09 Savings goals	F09	Implemented	`GoalService`; `CashFlowForecaster`; `GoalForecast`; `CreateGoalCommand`	`createGoal`, `forecast`	Command	D	A10, B7
F10 AI CFO chat	F10	Implemented	`CfoAgent`; `ToolRegistry`; `AbstractFinanceTool` + 12 tools; `PromptBuilder`; `ResponseValidator`; `PendingActionStore`; `ClaudeClientAdapter`, `MockLlmClient`, `LlmClientFactory`	`handle`, `runLoop`, `validate`, `execute`	Adapter, Template Method, Factory Method, Command	AI	A17, B11
F11 Can I afford it?	F11	Implemented (TIGHT verdict added)	`AffordabilityAnalyzer`; `AffordabilityResult`; `CashFlowForecaster`	`analyze`	-	D	A11, B7
F12 Monthly report	F12	Implemented	`ReportService`; `ReportSummarizer` (AI text checked against the figures, template fallback); `MonthlyReport`	`buildMonthlyReport`, `summarize`, `templateSummary`	-	H	A18, B12
F13 Auto-sync and reminders	F13	Modified (monthly check-in, choosable folder)	`ImportFolderWatcher`; `BalanceSyncService`; `ReminderService`; `SettingsRepository`	`start`, `processFile`, `accountFor`, `isMonthlyCheckinDue`	Observer	D	A1, B13
F14 Detected contributions and RRSP estimate	F14	Implemented	`ContributionDetector`; `RecordContributionCommand` (also moves money into the TFSA/RRSP/FHSA account); `RrspRoomEstimator`	`scan`, `detect`, `estimate(year, pa)`	Observer, Command	D	A2, A14, B9
F15 Self-updating limits	F15	Modified (GitHub issue instead of email)	`LimitsService`; `RemoteLimitsProvider`, `BundledLimitsProvider`, `LimitsCache`; `LimitsValidator`; `LimitsUpdateJob`, `CraLimitsScraper`, `GitHubIssueNotifier`, `ConsoleNotifier`; `.github/workflows/limits-update.yml`	`refreshIfStale`, `validate`, `run`	Strategy (providers)	D	A20, B14
F16 AI reads an unknown bank format	New	Implemented	`AiFormatReader` (AI proposes a mapping; Java builds and test-reads it); `AiFormatReader.Proposal`	`propose`, `parseJson`, `saveFormatAndImport`	Adapter	H	A4, B2

F17 AI monthly review	New	Implemented	`MonthlyReviewService`; `CfoAgent.runTask`; `BudgetPlanner`; `BudgetSuggestionTool`	`review`, `runTask`, `suggest`	Command, Template Method	AI	A19, B11
F18 Find transactions in plain English	New	Implemented	`TransactionQueryParser` (AI fills filter fields; Java checks them); `TransactionFilter`	`parse`, `fromJson`, `apply`, `describe`	-	H	A6, B4
F19 Where extra money could go	New	Implemented	`ExtraMoneyPlanner` (Java split); `ExtraMoneyTool`; `CfoAgent` explains	`plan`, `explainExtraMoney`	Template Method	H	A12, B8

Design-to-implementation traceability (Stage 1 classes → implemented files)

The "Designed classes" column is copied from the Stage 1 traceability table (Stage 1 report, Section 8). Every designed class exists under the same name in `src/main/java/com/maplecfo/` unless the last column says otherwise; the reasons for each change are in Section 4.

Feature	Designed classes (Stage 1)	Implemented classes / files	Status	Differences from the design
F01	TransactionsView, MapleCfoFacade, TransactionService, MappingProfileRepository, ColumnMapping, TransactionImporter, CsvTransactionAdapter, TransactionRepository	All designed classes, plus `DefaultProfiles`, `ImportResult`, `TransactionFilter`	Modified	Format detected automatically (`detectProfile`)
F02	CategorizationService, RuleBasedCategorizer, LlmCategorizer, LlmClient, PromptBuilder, RecategorizeCommand, CommandHistory, TransactionService	All designed classes, plus `DefaultRules`	Implemented	-
F03	DashboardView, MapleCfoFacade, AccountService, NetWorthService, AssetComponent, AccountGroup, Account, AccountRepository	All except DashboardView → `HomeView` + `AccountsView`	Implemented	Dashboard split into GUI pages
F04	BudgetView, BudgetService, SetBudgetCommand, CommandHistory, EventBus, AlertService, FinanceEventListener	All; BudgetView → `BudgetsView`	Implemented	Renamed view
F05	DashboardView, MapleCfoFacade, TransactionService, RecurringChargeDetector	All; DashboardView → `SubscriptionsView`; plus `RecurringCharge`	Modified	Price steps and cancelled charges
F06	PlanningView, RegisteredAccountService, ContributionLimits, ProfileRepository, ContributionRepository, EventBus	All; PlanningView → `RegisteredView`; plus `ContributionRoom`, `ContributionCheck`, `ContributionKind`, `RepaymentPlan`	Modified	HBP/LLP, FHSA transfer, RRSP allowance
F07	PlanningView, MapleCfoFacade, FireCalculator	All; PlanningView → `RetirementView`; plus `FireInputs`, `FireProjection`	Modified	Own target, target age, Coast FIRE
F08	PlanningView, DebtPayoffPlanner, PayoffStrategy, AvalancheStrategy, SnowballStrategy, DebtRepository	All; PlanningView → `DebtView`; plus `PayoffPlan`	Implemented	-
F09	GoalsView, GoalService, CreateGoalCommand, CommandHistory, CashFlowForecaster	All, plus `GoalForecast`	Modified	Surplus shared between goals
F10	ChatView, MapleCfoCli, CfoAgent, ConversationMemory, PromptBuilder, LlmClient, ClaudeClientAdapter, ToolRegistry, FinanceTool, AbstractFinanceTool, ResponseValidator, PendingActionStore	All, plus `LlmClientFactory`, `MockLlmClient`, `DemoRules`, 12 tool classes	Implemented	-
F11	GoalsView / ChatView, AffordabilityTool, AffordabilityAnalyzer, CashFlowForecaster, GoalService, BudgetService, ProposeActionTool, PendingActionStore, CommandHistory	All; the GUI page is `AffordView`; plus `AffordabilityResult`	Modified	TIGHT verdict added
F12	ReportView, MapleCfoCli, ReportService, ReportSummarizer, LlmClient, BudgetService, NetWorthService, RecurringChargeDetector	All, plus `MonthlyReport`	Implemented	-
F13	ImportFolderWatcher, TransactionService, BalanceSyncService, AccountService, DebtRepository, ReminderService, EventBus	All, plus `SettingsRepository`, `SettingsView`	Modified	User-chosen folder, monthly check-in
F14	ContributionDetector, RecordContributionCommand, PendingActionStore, CommandHistory, RegisteredAccountService, RrspRoomEstimator	All designed classes	Modified	Partial-year scaling, pension adjustment
F15	LimitsService, LimitsProvider, RemoteLimitsProvider, BundledLimitsProvider, LimitsValidator, LimitsCache, LimitsUpdateJob, CraLimitsScraper, FailureNotifier, GitHubIssueNotifier, EmailNotifier	All except FailureNotifier → `LimitsNotifier` and EmailNotifier → `ConsoleNotifier`	Modified	No e-mail; bot opens a pull request or an issue
F16-F19	New in Stage 2 (no Stage 1 classes)	See the mapping table above	New	UC16-UC19, SD15-SD18

F10-F19	New in Stage 2 (no Stage 1 classes)	See the mapping table above	New	UC10-UC19, SD13-SD18

How the classes collaborate (user input to result)

- **F01 Import.** The user picks a file (GUI) or runs `import file -a chequing` (CLI). `MapleCfoFacade.importTransactionsAuto` asks `TransactionService.detectProfile` which saved `ColumnMapping` fits the first row, then `importFrom` creates a `CsvTransactionAdapter` for that mapping (Adapter: any bank layout becomes `Transaction` objects). Each row gets a fingerprint; rows already stored are counted as duplicates. New rows are categorized (F02), saved through `TransactionRepository`, and a `TRANSACTIONS_IMPORTED` event is published, so budgets, balances and contribution detection update themselves (Observer). The `ImportResult` (imported, duplicates, rejected rows with reasons) is shown.
- **F02 Categorize.** `CategorizationService` runs its strategies in order: `RuleBasedCategorizer` (learned rules first, then built-in keywords; the longest keyword wins), then `LlmCategorizer`, which accepts the AI's answer only when it is exactly one category name. A correction runs a `RecategorizeCommand` through `CommandHistory` (undoable) and saves a learned rule.
- **F03 Net worth.** `AccountService.buildAccountTree` builds groups (cash, registered, other, debts) of `Account` leaves; `getValue` on the root sums the tree, with liabilities negative (Composite). `NetWorthService` stores snapshots.
- **F04 Budgets.** `SetBudgetCommand` saves a `Budget`. `BudgetService` listens to import and recategorize events, recomputes `BudgetStatus` and publishes `BUDGET_WARNING` (80%) and `BUDGET_EXCEEDED` (100%) once; `AlertService` turns events into messages.
- **F05 Recurring.** `RecurringChargeDetector` groups 12 months of expenses by normalised merchant, checks that the gaps fit weekly, monthly or yearly (plus or minus 3 days), flags a price increase (latest charge or a step from one stable price to a higher one), labels variable bills, rolls the next date forward and drops charges that missed two dates.
- **F06 Room.** `RegisteredAccountService.computeRoom` loads the profile and the CRA limits (`LimitsService`), then applies the rules for each account type (TFSA room since age 18 with withdrawals returned on January 1, RRSP limit from the Notice of Assessment with the $2,000 allowance, FHSA $8,000/year with carry-forward, HBP/LLP that use no room). The contribution form calls `previewContribution` on every keystroke, which has no side effects.
- **F07 FIRE.** `FireInputs` validates the inputs; `FireCalculator.project` computes the target (spending / withdrawal rate, or the user's own target), simulates each year, adds what-if scenarios, the Coast FIRE number and, when a retirement age is given, the yearly saving needed (annuity formula).
- **F08 Debt.** `DebtPayoffPlanner` orders debts with the chosen `PayoffStrategy` (avalanche: highest rate first; snowball: smallest balance first) and simulates month by month.
- **F09 Goals / F11 Afford.** `CashFlowForecaster` averages income, spending and registered transfers over the last complete months. `GoalService.forecast` shares the monthly surplus between goals; `AffordabilityAnalyzer` adds cash, expected savings and a one-month buffer and returns YES / TIGHT / NO / INSUFFICIENT_DATA.
- **F10 Chat.** `CfoAgent.handle` builds the prompt (`PromptBuilder`), sends it with the tool schemas through `LlmClient`, runs the requested tools through `ToolRegistry` (each tool validates its arguments in `AbstractFinanceTool.execute`, Template Method), and loops until the model answers. `ResponseValidator` checks every amount; if one is not in a tool result the model gets one correction round, otherwise the answer is flagged. A `propose_action` call stores a `SetBudgetCommand` in `PendingActionStore` until the user confirms.
- **F12 Report.** `ReportService` computes income, spending by category, budgets, recurring costs and price increases for a month; `ReportSummarizer` asks the AI for a short summary and accepts it only if its numbers match the report, otherwise it uses a template.
- **F13 Auto-sync.** `ImportFolderWatcher` (Java `WatchService`) waits until a new file stops growing, detects its format, picks the account from the file name and imports it. `BalanceSyncService` reads the running balance and updates the account and linked debt. `ReminderService` raises the monthly check-in and "no import in 30 days" reminders.
- **F14 Contributions.** `ContributionDetector` listens to imports and proposes a `RecordContributionCommand` for each transfer that names exactly one of TFSA/RRSP/FHSA; on confirmation the room and the account balance both change. `RrspRoomEstimator` estimates RRSP room from a year of pay deposits (scaled for a partial year, minus the pension adjustment).
- **F15 Limits.** On start `LimitsService.refreshIfStale` tries `RemoteLimitsProvider` (GitHub, at most weekly), checks the data with `LimitsValidator` and caches it (`LimitsCache`), falling back to the bundled JSON. The weekly GitHub Actions job runs `LimitsUpdateJob`, which scrapes the CRA pages, validates the numbers and opens a pull request or, on a problem, an issue (`GitHubIssueNotifier`).
- **F16 AI format reader.** When no format fits, `AiFormatReader.propose` sends the first 8 lines to the model, turns the JSON reply into a `ColumnMapping` (invalid answers are rejected), and test-reads the whole file with `CsvTransactionAdapter`. The user sees a preview and the counts of rows read and rejected; only then is the format saved and the file imported (`saveFormatAndImport`).
- **F17 Monthly review.** `MonthlyReviewService.review` gives `CfoAgent.runTask` an instruction; the agent calls `spending_summary`, `budget_status`, `recurring_charges`, `goal_forecast`, `contribution_room` and `suggest_budgets` (`BudgetPlanner`: three-month average rounded up to $10), proposes up to five budgets with `propose_action`, and writes a to-do list. Each proposal is confirmed or dismissed separately (see SD15).
- **F18 Plain-English search.** `TransactionQueryParser.parse` asks the model to fill the fields of a `TransactionFilter`; `fromJson` checks each one (real dates, known accounts and categories, sensible amounts) and lists what it ignored. The GUI shows the result in the normal filter controls, so the user can see and change it.
- **F19 Extra money.** `ExtraMoneyPlanner.plan` splits the amount in a fixed educational order (debts at 8% or more, an emergency fund of three months of spending, FHSA, TFSA, RRSP, cheaper debt) using the user's own debts, cash, spending and room. The GUI shows the split at once; `explainExtraMoney` asks the agent to explain it through `ExtraMoneyTool`.

3. Implemented design patterns

All eight patterns from the Stage 1 design are implemented; Repository is used in addition. Each one solves a concrete problem in the code.

3.1 Facade: `MapleCfoFacade`

- **Problem.** Two user interfaces (GUI and CLI) need about 25 services; without a single entry point each screen and command would wire services itself and the two interfaces would drift apart.
- **Participants.** Facade `MapleCfoFacade`; subsystems `TransactionService`, `BudgetService`, `RegisteredAccountService`, `CfoAgent`, `AiFormatReader`, `MonthlyReviewService`, etc.; clients: the 13 GUI views and the CLI command classes. `AppContext` (composition root) builds everything once.
- **Code.** `app/MapleCfoFacade.java` (e.g. `importTransactionsAuto`, `askCfo`, `runMonthlyReview`, `planExtraMoney`).
- **Why it fits.** Both interfaces call the same methods, so a feature built once works in both; tests use the facade end to end. **Matches Stage 1**; it gained the methods for F16-F19.

3.2 Adapter: `CsvTransactionAdapter`, `ClaudeClientAdapter`

- **Problem.** Every bank exports a different CSV layout, and the Anthropic HTTP API has its own message format; the rest of the code should see only `Transaction` objects and a simple `LlmClient`.
- **Participants.** Target `TransactionImporter` / `LlmClient`; adapters `CsvTransactionAdapter` (configured by a `ColumnMapping`) and `ClaudeClientAdapter` (wraps `HttpTransport`); `MockLlmClient` is a second `LlmClient`.
- **Code.** `importer/CsvTransactionAdapter.importFile`, `agent/llm/ClaudeClientAdapter.chat`.
- **Why it fits.** A new bank is a new mapping, not new code (F16 even lets the AI propose the mapping); the agent can run on Claude, on the offline mock or on a scripted fake in tests. **Matches Stage 1.**

3.3 Strategy: `PayoffStrategy`, `CategorizationStrategy`, `LimitsProvider`

- **Problem.** Debts can be paid in different orders; transactions can be categorized by rules or by the AI; limits can come from several sources. The algorithms must be swappable without `if/else` chains.
- **Participants.** `PayoffStrategy` with `AvalancheStrategy` and `SnowballStrategy` (context `DebtPayoffPlanner`); `CategorizationStrategy` with `RuleBasedCategorizer` and `LlmCategorizer` (context `CategorizationService`, an ordered chain); `LimitsProvider` with remote, cache and bundled providers (context `LimitsService`, ordered fallback).
- **Code.** `planning/DebtPayoffPlanner.plan`, `categorization/CategorizationService.categorizeAll`, `LimitsService.reload`.
- **Why it fits.** The GUI switches avalanche/snowball at run time; the AI categorizer is just a second strategy that is skipped in demo mode. **Matches Stage 1**; the limits providers are an extra use.

3.4 Observer: `EventBus` and `FinanceEventListener`

- **Problem.** After an import many things must update (budgets, alerts, balances, detected contributions, the GUI) without `TransactionService` knowing about all of them.
- **Participants.** Subject `EventBus` (`publish`, `subscribe`); event `FinanceEvent` with `EventType`; observers `BudgetService`, `AlertService`, `BalanceSyncService`, `ContributionDetector`, and the GUI (`MapleCfoApp` refreshes).
- **Code.** `events/EventBus.publish`, `BudgetService.onEvent`, `ContributionDetector.onEvent`.
- **Why it fits.** F13 and F14 were added as new listeners without changing the import code; a failing listener does not stop the others. **Matches Stage 1**; new event types `BUDGET_WARNING` and `AUTO_IMPORTED`.

3.5 Command: `UndoableCommand`, `CommandHistory`, `PendingActionStore`

- **Problem.** Edits must be undoable, and changes suggested by the AI or by automation must not run until the user agrees.
- **Participants.** `UndoableCommand` (`execute`, `undo`, `describe`); concrete `SetBudgetCommand`, `RecategorizeCommand`, `CreateGoalCommand`, `RecordContributionCommand`; invoker `CommandHistory` (undo/redo stacks); `PendingActionStore` holds proposed commands until `confirmProposedAction`.
- **Code.** `command/CommandHistory.run/undo/redo`, `agent/tools/ProposeActionTool`, `MapleCfoFacade.confirmProposedAction`.
- **Why it fits.** One mechanism gives Undo/Redo in both interfaces and a safe “propose, then confirm” step for F10, F14 and F17. **Matches Stage 1**; used more widely (contributions, monthly review).

3.6 Composite: `AssetComponent`, `Account`, `AccountGroup`

- **Problem.** Net worth is a tree (cash, registered, debts, each with accounts) and should be computed the same way at every level.
- **Participants.** Component `AssetComponent` (`getValue`, `getChildren`); leaf `Account`; composite `AccountGroup`.
- **Code.** `domain/AccountGroup.getValue`, `planning/AccountService.buildAccountTree`.
- **Why it fits.** Liabilities return negative values, so every total is a plain sum. **Matches Stage 1.**

3.7 Template Method: `AbstractFinanceTool`

- **Problem.** All 12 agent tools must validate the model’s arguments and turn failures into errors the model can read; no tool may skip this.
- **Participants.** Abstract class `AbstractFinanceTool` with the final template `execute` (validate the schema, call `doExecute`, catch failures); concrete tools such as `AffordabilityTool`, `ExtraMoneyTool`, `BudgetSuggestionTool`.
- **Code.** `agent/AbstractFinanceTool.execute`, `validateArgs`, `doExecute`.
- **Why it fits.** The two new tools for F17 and F19 only wrote `doExecute`. **Matches Stage 1.**

3.8 Factory Method: `LlmClientFactory`

- **Problem.** The app must choose the real Claude client or the offline mock from the configuration.
- **Participants.** `LlmClientFactory.create(AppConfig, rules)` returns an `LlmClient`.
- **Why it fits.** No other class knows whether a key exists. **Matches Stage 1.**

3.9 Repository (additional)

`Repository<T, ID>` and its interfaces (transactions, accounts, budgets, goals, debts, profile, contributions, mapping profiles, settings) with SQLite implementations and in-memory versions for tests. Services never see SQL. `DatabaseManager` migrates old databases (adds new columns, keeps data).

3.10 Updated UML

The class diagram was updated to the implemented system in UMLet (8 parts), the use-case diagram gained UC16-UC19, SD13 was corrected, and SD15-SD18 were added for the new features. All diagrams (editable `.uxf` and `.png`) are in `docs/stage2/diagrams`; the class diagrams are shown in Appendix A and the new sequence diagrams in Appendix B.

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4. Changes and rationale

Original design → implemented change → reason → impact.

Area	Original design (Stage 1)	Implemented change	Reason	Impact
New feature	-	**F16** AI reads an unknown bank format (`AiFormatReader`, UC16, SD16)	Users of other banks had to map columns by hand	Any bank's CSV can be imported after one check of the preview; uses Adapter (new `ColumnMapping`)
New feature	-	**F17** AI monthly review with proposed budgets (`MonthlyReviewService`, `BudgetPlanner`, `CfoAgent.runTask`, UC17, SD15)	Show real multi-step agent planning; turn the monthly check-in into concrete actions	`AgentResponse` now carries several proposals; Command + PendingActionStore reused
New feature	-	**F18** plain-English transaction search (`TransactionQueryParser`, `TransactionFilter`, UC18, SD17)	Filters are powerful but slow to set	AI only fills checked fields; search stays deterministic
New feature	-	**F19** where to put extra money (`ExtraMoneyPlanner`, `ExtraMoneyTool`, UC19, SD18)	A common question after a tax refund or bonus	New tool via Template Method; amounts from Java
F01	User picks a mapping profile on every import	Format detected from the columns (`detectProfile`, `matchesFirstRow`); choice optional; transaction filters added	Found while testing the GUI: picking the bank every time was confusing	Used by F01, F13 and F16
F05	Latest charge vs. average; 6-month window	12-month window, price-step detection, cancelled charges dropped, new price used for yearly cost	The 22-month demo data showed price changes labelled "amount varies" and cancelled subscriptions still listed	More accurate subscription page and report
F06	Contribution or withdrawal; anything over the limit is over-contributed	RRSP $2,000 allowance; same-year withdrawal clears an excess; HBP/LLP withdrawals and repayments; FHSA→RRSP transfer; pension adjustment	Checked against CRA rules after testing	`ContributionKind`, `RepaymentPlan`, DB migration
F06/F03	Confirmed contributions change only room	They also move money into the TFSA/RRSP/FHSA account; forecasts subtract registered transfers	Net worth dropped after a contribution (found in testing)	`RecordContributionCommand`, `CashFlowForecast.cashSurplus`
F07	Fixed 5% return, 4% rule	Own target, target age with saving needed, Coast FIRE, editable return/withdrawal rate, "use my numbers", remembered plan	Requested during review	`FireInputs`/`FireProjection` fields; tool accepts the new arguments
F11	YES / NO	YES / TIGHT / NO / INSUFFICIENT_DATA	"Affordable only by using the safety buffer" needed its own answer	GUI and agent show the verdict
F09	Each goal forecast alone	Monthly surplus shared between goals	Several goals could each claim the same money	More realistic forecasts
F13	Fixed inbox folder; 30-day reminder	User-chosen folder (Settings); monthly check-in; account from the file name	Usability	`SettingsRepository`, `ReminderService.isMonthlyCheckinDue`
F14	RRSP estimate from last year's income	Partial year scaled to 12 months; pension adjustment subtracted; labelled as a low estimate	Most users import only part of a year	`RrspRoomEstimator.estimate(year, pa)`
F15 (pattern detail)	`EmailNotifier` sends failure emails	`GitHubIssueNotifier` + `ConsoleNotifier`; bot opens a pull request instead of committing	An SMTP password in CI is a security risk; a PR lets a human review new limits	SD13 updated; no secrets needed
GUI	Tabbed main window with 7 views	Sidebar with 13 pages, Home checklist, in-window pop-ups (AtlantaFX `ModalPane`), live checks	Krishna found the first GUI hard to use and dated	Class diagram 6/8 redrawn
Agent	One proposal per answer, 6 steps	`runTask` with more steps for long tasks; list of proposals; memory not used for tasks	Needed by F17	Class diagram 5/8
Persistence	Schema fixed	`DatabaseManager` adds new columns to old databases	New fields (pension adjustment) must not break existing data	Migration test

No Stage 1 feature or pattern was removed.

5. AI tools and collaboration log

5.1 Tools and models

Tool / platform	LLM / model	Purpose and example tasks	Total tokens	Money cost
Claude (Anthropic), Claude Pro plan, single user	claude-opus-5-5 (configured)	Writing all Java code and JUnit tests from the Stage 1 design (all 19 features, 8 patterns, GUI, CLI), bug diagnosis, demo dataset, README, UML updates, recording helpers for the videos, this report draft	Not shown per message on a subscription. Measured from the session log for Nov 2-13: about 0.33 million output tokens, 5.0 million tokens written to the prompt cache and 131 million read from it (repeated context). Earlier sessions (Oct 20-Nov 1) are not available	Flat subscription: Claude Pro, US$20 a month (list price with monthly billing), no per-token charge
Anthropic Claude API (inside MapleCFO, my own key `maple-cfo-local`)	claude-haiku-5-5	Runtime LLM for F02 categorization, F10 chat, F12 summary, F16-F19, used while testing and recording the videos	478,875 tokens (Nov 3-12, from the Anthropic Console usage export): 429,296 input, 49,579 output, no cache	About US$0.07: 429,296 x $0.10 + 49,579 x $0.50 per million tokens (Haiku 5.5 list prices)

About the model name. The Claude app was set to `claude-opus-5-5`, but the app does not show which model served each individual reply, and it can switch models (for example as a fallback). So `claude-opus-5-5` is the configured model, not a guarantee for every interaction. The model used inside MapleCFO is exact, because the app sends it with every API call (`LLM_MODEL`, default `claude-haiku-5-5`).

The app itself shows the tokens used in a session (`ai` command in the CLI), which helped estimate the runtime cost.

5.2 How the collaboration worked

Following the course rule, the code was written by the AI from the Stage 1 design; I gave the instructions, reviewed each change, made the design decisions, ran the tests and the application on my own computer, reported defects, and made every commit. Each change came as one patch per commit; I applied it, ran `mvn test` and the app, and pushed only when it worked. I tested each feature as it was built. When a feature did not pass (a wrong figure, a confusing screen, an error on my Mac), we worked on it together and changed it until it did. I also asked for UI and UX changes to make the app easier for the end user: a sidebar with grouped pages instead of tabs, a Home page with a check-in list and “Needs your attention”, pop-ups inside the window, live checks while typing (for example the over-contribution warning), plain-English labels and messages, and the option to ask questions in plain language.

The full log (40 entries) is in `docs/stage2/AI_Collaboration_Log.md`.

5.3 Representative interactions

1. Protecting the API key

- *Task:* set up the Claude API key without exposing it.
- *Input:* “Can you check my file if I did it right?”
- *AI contribution:* checked the local files without printing the key; found the key had been typed into `.env.example` (a committed file) instead of `.env`; confirmed it was never committed or pushed; restored the placeholder and verified `.env` is git-ignored.
- *Human contribution:* I created a new key, put it only in `.env`, and revoked the exposed key in the Anthropic Console.
- *Outcome:* no key in the repository or its history; the handout’s key-protection rules are followed.

2. Testing the CLI found two F05 defects

- *Task:* run the built application on real use and check the results.
- *Input:* I ran `--help`, `import`, `recurring` and `fire` on my Mac and shared the output.
- *AI contribution:* found a next-expected date in the past and a “price went up” flag on a variable utility bill.
- *Human contribution:* I ran the commands on my own Mac and shared the real output; I noticed the odd dates and flags in the list and asked for them to be checked.
- *Outcome:* next dates now roll forward and variable bills are not flagged (commit `e832174`).

3. F10 AI CFO agent

- *Task:* implement the bounded tool-calling agent from SD08.
- *Input:* “Implement the agent with grounding checks and confirm-before-change.”
- *AI contribution:* `CfoAgent` loop, `ToolRegistry`, `AbstractFinanceTool` (Template Method), tools, `ResponseValidator`, `PendingActionStore`, CLI `ask/confirm/dismiss`.
- *Human contribution:* ran the tests, then asked “How much did I spend on dining last month?” with my own data; the answer ($162.04) matched my bank data.
- *Outcome:* committed `439f701`; answers are grounded in tool results.

4. Rent counted as a transfer

- *Task:* AI categorizer and monthly report.
- *AI contribution:* `LlmCategorizer`, `ReportService`, `ReportSummarizer` with figure checking.
- *Human contribution:* tested `report -m 2026-09` and found September spending was far too low ($921.61 instead of $2,371.61) because rent paid by e-transfer was categorised as TRANSFER.
- *Outcome:* transfer rules now run last, so “rent” wins; re-imported on a fresh database (`e494302`, `439f701`).

5. Automatic bank-format detection (my idea)

- *Task:* stop asking the user to pick the CSV format.
- *Input:* “Instead of selecting the CSV type, should we make it dynamic?”
- *AI contribution:* detection from the file’s columns in the GUI and CLI, with a clear message when nothing matches.
- *Human contribution:* suggested the change while testing the GUI.
- *Outcome:* `8535411`; later reused by auto-import and by F16.

6. GUI usability rework

- *Task:* make the GUI easy to use.
- *Input:* “The GUI is not user friendly, hard to navigate and still looks old-fashioned”; I listed the problems (too many fields, nowhere to start, unclear where to enter what) and chose a full rework over a restyle.
- *AI contribution:* 13 pages in a sidebar, Home checklist, short forms in pop-ups, empty states, AtlantaFX theme.
- *Human contribution:* rejected two earlier GUI versions, found overlapping report labels on my laptop, and found that pop-ups opened as separate windows on macOS.
- *Outcome:* `63a6ed7`, `60d6ae5`, `921f649`.

7. Following the money for TFSA contributions

- *Task:* check what happens to a TFSA contribution.
- *Input:* “Is the TFSA contribution being added to the net worth or other things?”
- *AI contribution:* confirmed contributions now also update the TFSA/RRSP/FHSA account and forecasts subtract regular registered transfers; monthly check-in and a choosable import folder.
- *Human contribution:* found the gap (net worth dropped after a contribution) by asking how the money flows; asked for the monthly reminder and folder setting.
- *Outcome:* `ada881a`.

8. A test that depended on my computer

- *AI contribution:* two tests read `MAPLECFO_INBOX` from the environment; settings now come from `AppConfig` and tests use empty settings.
- *Human contribution:* `mvn test` failed on my Mac; I reported the output.
- *Outcome:* tests no longer depend on the machine (`921f649`).

9. TFSA stayed negative after a withdrawal

- *AI contribution:* room only counted withdrawals from earlier years; a same-year withdrawal now clears an excess (but adds no new room), matching CRA rules.
- *Human contribution:* found the bug by over-contributing and then withdrawing in the GUI.
- *Outcome:* `3ec76b8`, 3 new tests.

10. Real CRA rules (HBP, LLP, allowance)

- *Input:* “How does the limit work in real life?”, then “let’s implement this”.
- *AI contribution:* explained the rules, then implemented the RRSP $2,000 allowance, HBP/LLP withdrawals and repayments, FHSA-to-RRSP transfers and the pension adjustment, with a database migration.
- *Human contribution:* asked how the limits work in real life, asked for the real rules to be implemented, and reviewed the result in the GUI (over-contribution warning, HBP repayment banner).
- *Decision:* LLP repayment start simplified (noted in limitations).
- *Outcome:* `6f536e4`, `a985778`.

11. Realistic test data exposed detector gaps

- *Input:* “Can we add more CSV files so there is a good dataset for testing and the videos?”
- *AI contribution:* a 22-month fake dataset with a load script and `DemoDatasetTest`; the data showed that price changes from months ago were labelled “amount varies” and cancelled subscriptions were still listed; both fixed.
- *Human contribution:* asked for the dataset, loaded it on my Mac and used it for the real-key test and both demonstration videos.
- *Outcome:* `f7058dd`.

12. New AI features F16-F19 and FIRE options

- *Input:* “Can we introduce more AI features or use AI to drive things?”; later “in FIRE can we change the target”.
- *AI contribution:* proposed four features; I chose all four. Each keeps numbers in Java and asks before changing anything. FIRE gained an own target, a target age, Coast FIRE and editable assumptions.
- *Human contribution:* chose the features; on Nov 7 I ran F16-F19, F10 and F12 with my own API key and shared the output, which found the F17 wording defect (Section 6.3).
- *Outcome:* `f76b1dc`, `1b9f20b`, `5ef8658`, `f9070e6`, `4666a5d`.

What I rejected or corrected. I rejected the first two GUI designs; reported every failing build and test on my machine (including the iCloud duplicate-file problem, which led me to move the project out of iCloud); asked for real CRA rules instead of the simplified Stage 1 ones; and checked that every amount the agent showed matched my data.

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6. Development and verification summary

6.1 Weekly commit summary

Only commits that change implementation or test code are counted (documentation-only commits such as the README, the AI log and the UML are not). Every commit was pushed during its period.

Period	Code commits	Minimum
Period 1: Oct 20-26	5	5
Period 2: Oct 27-Nov 2	7	5
Period 3: Nov 3-9	28	5

Period 1: Oct 20-26

Commit	Date	Change
`0ca6048`	Oct 23	Maven project and Money value object with tests
`a2ff9b2`	Oct 23	Domain model: accounts (Composite), transactions, budgets, goals, debts
`5051cb3`	Oct 23	F07 FIRE calculator
`a369c1b`	Oct 23	F08 debt payoff planner (Strategy)
`db01f84`	Oct 23	F05 recurring charge detector

Period 2: Oct 27-Nov 2

Commit	Date	Change
`7c29fe5`	Oct 28	F06 room with CRA limits; EventBus (Observer)
`1904991`	Oct 28	F04 budgets with 80%/100% alerts
`4e43342`	Oct 28	Command pattern: undo/redo
`395b727`	Oct 29	F02 rule-based categorization
`7643cbb`	Nov 1	F01 CSV import, duplicates, sample data
`0c503e2`	Nov 1	SQLite repositories
`0d2ae51`	Nov 1	Facade and picocli CLI

Period 3: Nov 3-9

Commit	Date	Change
`e832174`	Nov 3	LLM layer (Claude adapter, mock, factory) + F05 fixes
`439f701`	Nov 3	F10 AI CFO agent
`e494302`	Nov 3	F02 AI categorizer, F12 report
`c4d5af8`	Nov 3	F03 net worth, F09 goals
`b925d3a`	Nov 6	F11 Can I afford it?
`06d505b`	Nov 6	JavaFX GUI part 1
`fd9a000`	Nov 6	GUI part 2 (planning, chat, report)
`05e7ee7`	Nov 6	F13 auto-import, F14 contribution detection
`fc1ab76`	Nov 6	F15 limits refresh and weekly bot
`5f45d8a`	Nov 6	More unit tests
`63a6ed7`	Nov 6	GUI redesign
`8535411`	Nov 6	Automatic bank-format detection
`60d6ae5`	Nov 6	GUI UX rework
`ada881a`	Nov 6	Registered transfers follow the money, monthly check-in
`921f649`	Nov 6	In-window pop-ups, live room check
`6f536e4`	Nov 6	Account activity, RRSP allowance
`3ec76b8`	Nov 6	Withdrawals clear an over-contribution
`a985778`	Nov 6	HBP/LLP, FHSA transfer, pension adjustment, filters
`f7058dd`	Nov 6	Demo dataset, recurring price steps
`f76b1dc`	Nov 6	F16 AI bank-format reader
`1b9f20b`	Nov 6	F18 plain-English search
`5ef8658`	Nov 6	F19 extra money
`f9070e6`	Nov 6	F17 AI monthly review
`4666a5d`	Nov 6	F07 FIRE options
`e4aff8a`	Nov 7	Logo: app icon and sidebar
`8eb9bb2`	Nov 7	Logo removed again: icons, sidebar image, class diagram 6
`560cc56`	Nov 7	F17 fix: review misread a next charge date as a price rise
`5f3de3d`	Nov 7	F05 fix: transfers no longer listed as recurring bills (+ test)

6.2 Checks performed

- **Automated tests:** 280 JUnit 5 tests in 32 classes, all passing (`mvn test`). They cover the domain rules, every calculator with exact expected values (FIRE, debt, room, affordability, goals), import (duplicates, bad rows, format detection), categorization, events and undo/redo, SQLite repositories and migration, every agent tool and its argument validation, the agent loop with scripted model replies (tool use, grounding check, correction round, proposals), the AI features F16-F19 with a fake model (valid, invalid and unreadable AI answers), CLI commands end to end, and the 22-month demo dataset (`DemoDatasetTest`).
- **Coverage (JaCoCo 0.8.12):** 59.7% of lines overall and 89.7% outside the JavaFX GUI package.
- **Running the application:** after each patch I ran `mvn test`, the CLI commands and the GUI on my Mac with the fake sample and demo data, and on Nov 7 ran F02, F10, F12 and F16-F19 with my real API key (this found the F17 defect below). Both demonstration videos were recorded with the real key.
- **GUI screenshots:** every page and pop-up was rendered headlessly (with and without data, at small and large window sizes) to catch layout problems before each commit.
- **Security checks:** `.env` and real bank CSVs are git-ignored, `.env.example` holds only a placeholder, and no key is in the repository or its history.

Full testing and the KUMA agent validation follow in Stage 3.

6.3 Important defects found and corrected

Defect	Found by	Fix
Recurring charges showed a next date in the past; variable bills flagged as price increases	Running the CLI on sample data	Roll dates forward; only fixed-price charges flagged (`e832174`)
Rent paid by e-transfer counted as a transfer, so spending was far too low	Testing the monthly report	Transfer rules run last (`439f701`)
Folder-watcher test could finish while an import was still writing	Coverage work	Test waits; `close()` waits for a running import (`5f45d8a`)
Report pie labels overlapped and text was cut off on a laptop screen	Using the GUI	Legend list beside the pie, responsive layout (`60d6ae5`)
Two tests depended on an environment variable on my machine	`mvn test` on my Mac	Settings read through `AppConfig`; tests use empty settings (`921f649`)
Dialogs opened as separate windows on macOS	Using the GUI	In-window pop-ups (`921f649`)
A confirmed TFSA contribution lowered net worth	Asking how the money flows	Contribution also moves money into the account (`ada881a`)
TFSA stayed over-contributed after a withdrawal	Using the GUI	Same-year withdrawal clears the excess (`3ec76b8`)
Price changes from months ago showed as “amount varies”; cancelled subscriptions listed	22-month demo data	Price-step detection, 12-month window, stopped charges dropped (`f7058dd`)
iCloud Desktop created “Name 2.java” copies and changed build output	Failing builds on my Mac	Never committed; project moved out of iCloud
AI monthly review said “Rent rises to $1,525 on 2026-12-01”, but the rent had already gone up (it read the next charge date as the date of the increase)	Testing F17 with my real API key	`recurring_charges` now says in words that the increase already happened; the review instruction says a next charge date is not a price change (`560cc56`)
Card payments, savings transfers and TFSA/FHSA contributions were listed as monthly bills, doubling the yearly total ($49,409 instead of $28,483)	Recording the CLI demo video	Transfers between own accounts are skipped; new test `ignoresTransfersBetweenOwnAccounts` (`5f3de3d`)

6.4 Remaining limitations

- Bank data comes only from CSV exports (no live bank connection, by design). TD, RBC and a generic layout are built in; other banks need F16 (AI key) or a manual mapping.
- AI features need an Anthropic API key. Without it, F10, F12, F17-F19 use clearly labelled fixed rules and F16 asks the user to choose the format.
- Tax rules are simplified for education: the RRSP estimate uses after-tax pay (so it is low), LLP repayments start in the 5th year, provincial differences and tax refunds are not calculated, and nothing is tax or investment advice.
- Proposals waiting for confirmation and the undo history are kept in memory, so in the CLI they work across commands only in interactive mode (each one-shot command is a new process).
- Recurring detection needs at least 3 charges, so yearly charges appear only after three years of data.
- Net-worth history records a snapshot only when a balance changes.
- The JavaFX GUI is tested by running it and by screenshots, not by unit tests (hence the lower overall coverage).
- The weekly limits bot depends on the CRA page layout; if it changes, the validator rejects the data and the bot opens an issue instead of updating.
- Single user and a local, unencrypted SQLite file.

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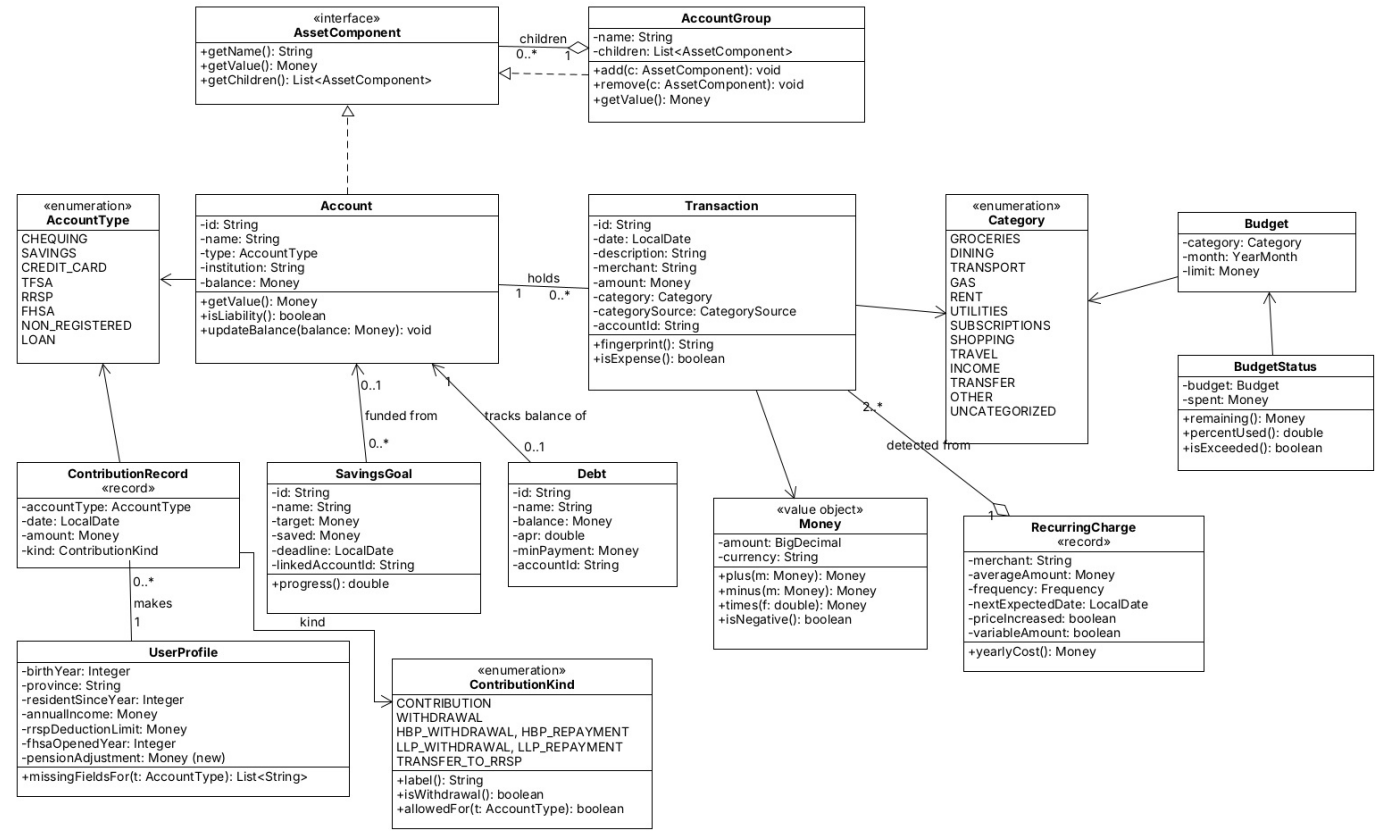
7. Demonstration index

Video A (GUI, 6:25): https://youtu.be/VF4bBqEF96I · Video B (CLI, 4:10): https://youtu.be/kkj3KK_1WIg · Both are unlisted on YouTube and open without an access request. Each link below starts at the feature's segment. Every segment shows the feature ID and name, how it is started, the input, the result, and a caption saying what it does and where the AI is involved; error cases are included where they exist (A7, B5, B10). Both videos were recorded on my Mac with the fake demo data and my real API key (the key is never shown). Both are screen recordings of my Mac. The GUI tour clicks through the app by itself (a recording helper in a separate file, not part of the app); in the CLI video a helper script types each command in my Terminal.

Feature	GUI (Video A)	CLI (Video B)
F01 Import bank CSVs	A3 0:49	B1 0:09
F02 Categorize and correct	A5 1:34	B3 0:41
F03 Accounts and net worth	A9 2:44	B7 1:28
F04 Budgets and alerts	A7 2:09	B5 1:04
F05 Recurring charges	A8 2:34	B6 1:21
F06 TFSA/RRSP/FHSA room	A13 3:41	B9 1:54
F07 FIRE projection	A16 4:24	B10 2:20
F08 Debt payoff	A15 4:11	B10 2:20
F09 Savings goals	A10 2:58	B7 1:28
F10 AI CFO chat	A17 4:48	B11 2:39
F11 Can I afford it?	A11 3:06	B7 1:28
F12 Monthly report	A18 5:22	B12 3:13
F13 Auto-sync and reminders	A1 0:09	B13 3:24
F14 Detected contributions, RRSP estimate	A2 0:27, A14 4:02	B9 1:54
F15 Self-updating limits	A20 6:04	B14 3:46
F16 AI reads an unknown bank format	A4 1:12	B2 0:19
F17 AI monthly review	A19 5:36	B11 2:39
F18 Find transactions in plain English	A6 1:53	B4 0:54
F19 Where extra money could go	A12 3:27	B8 1:44
Demo mode (no API key)	-	B15 3:58

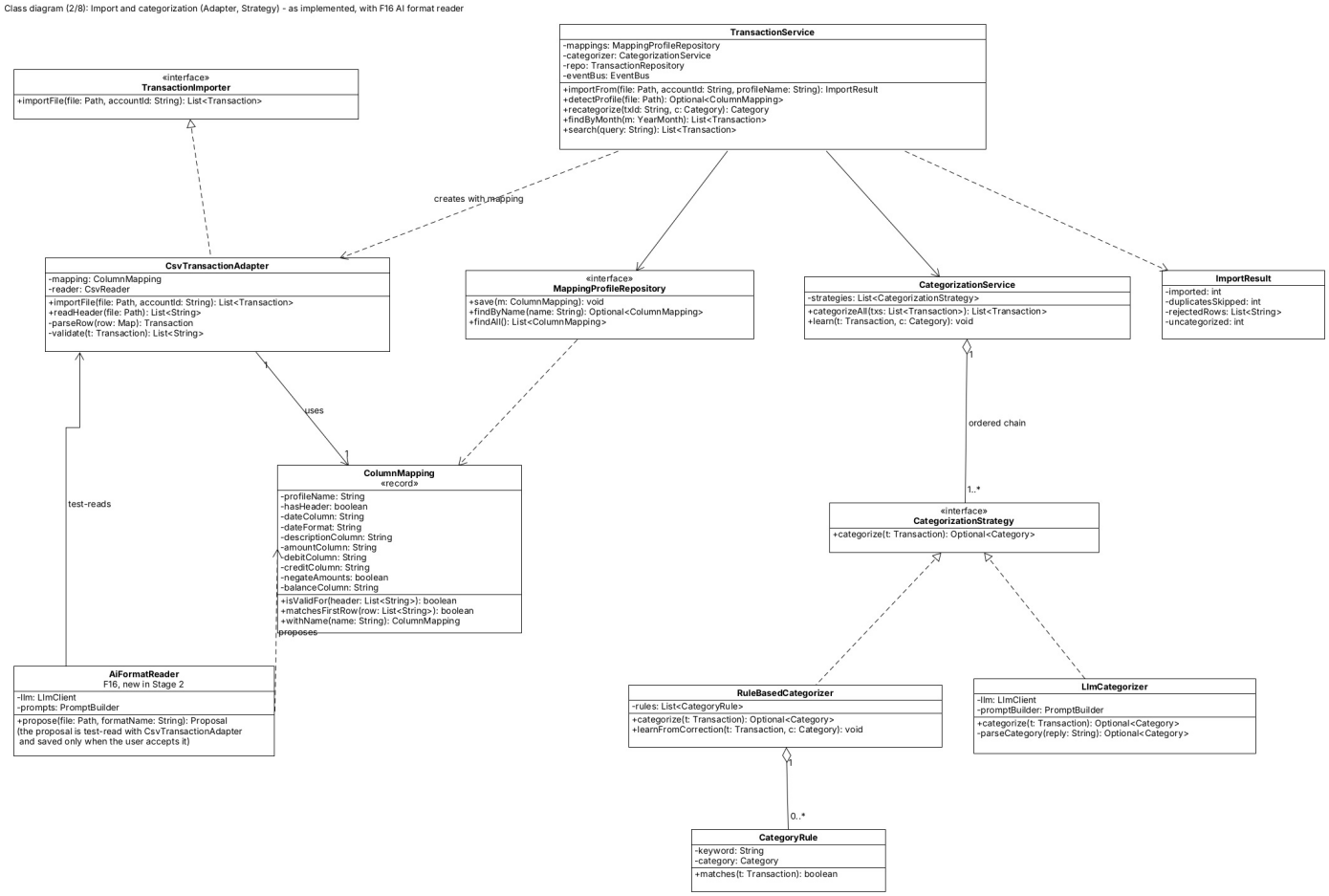
Appendix A: Updated class diagrams (Stage 2, as implemented)

Class diagram (1/8): Domain model (Composite: net worth) - as implemented in Stage 2



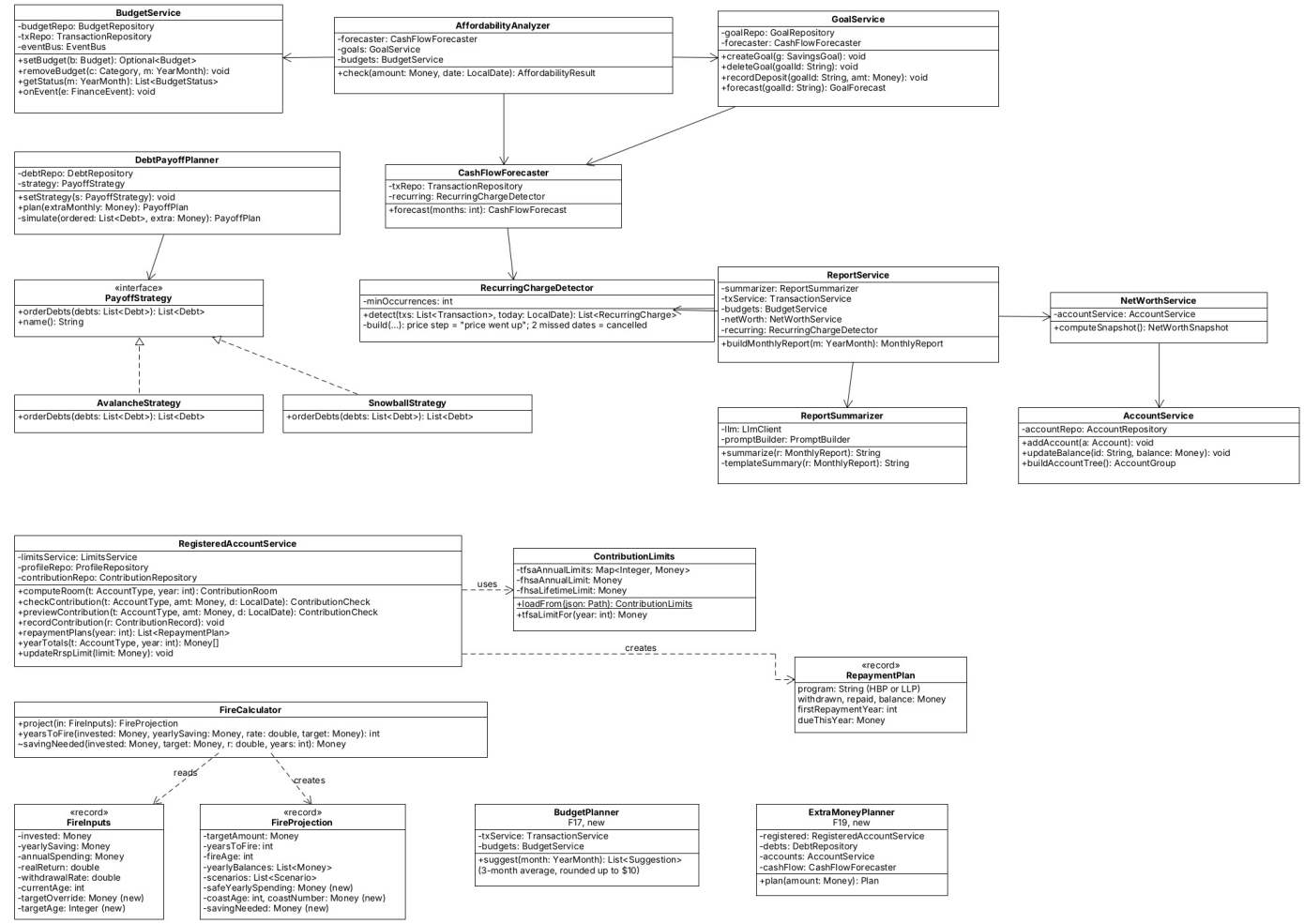
Class diagram 1/8

Class diagram (2/8): Import and categorization (Adapter, Strategy) - as implemented, with F16 AI format reader



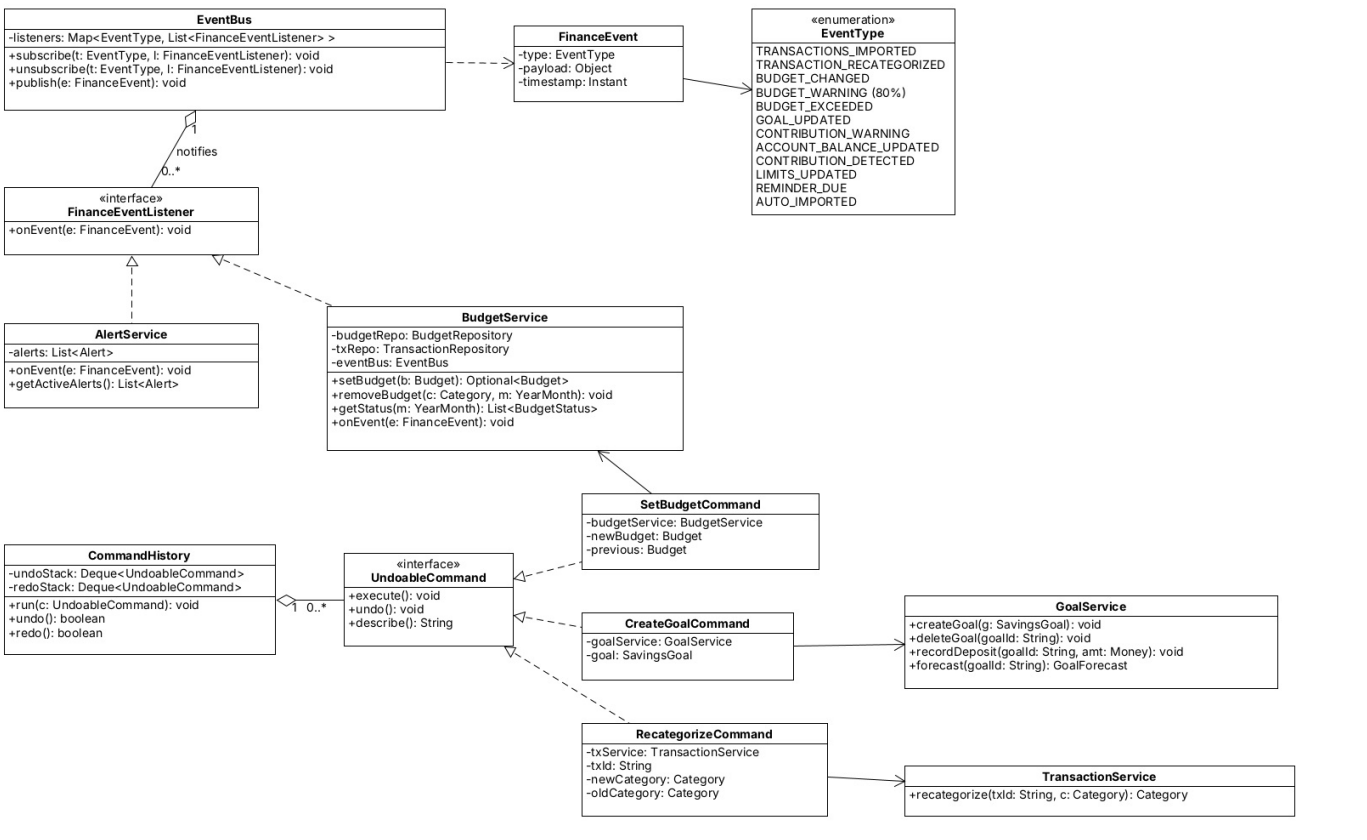
Class diagram 2/8

Class diagram (3/8): Planning and analysis services (Strategy) - as implemented in Stage 2

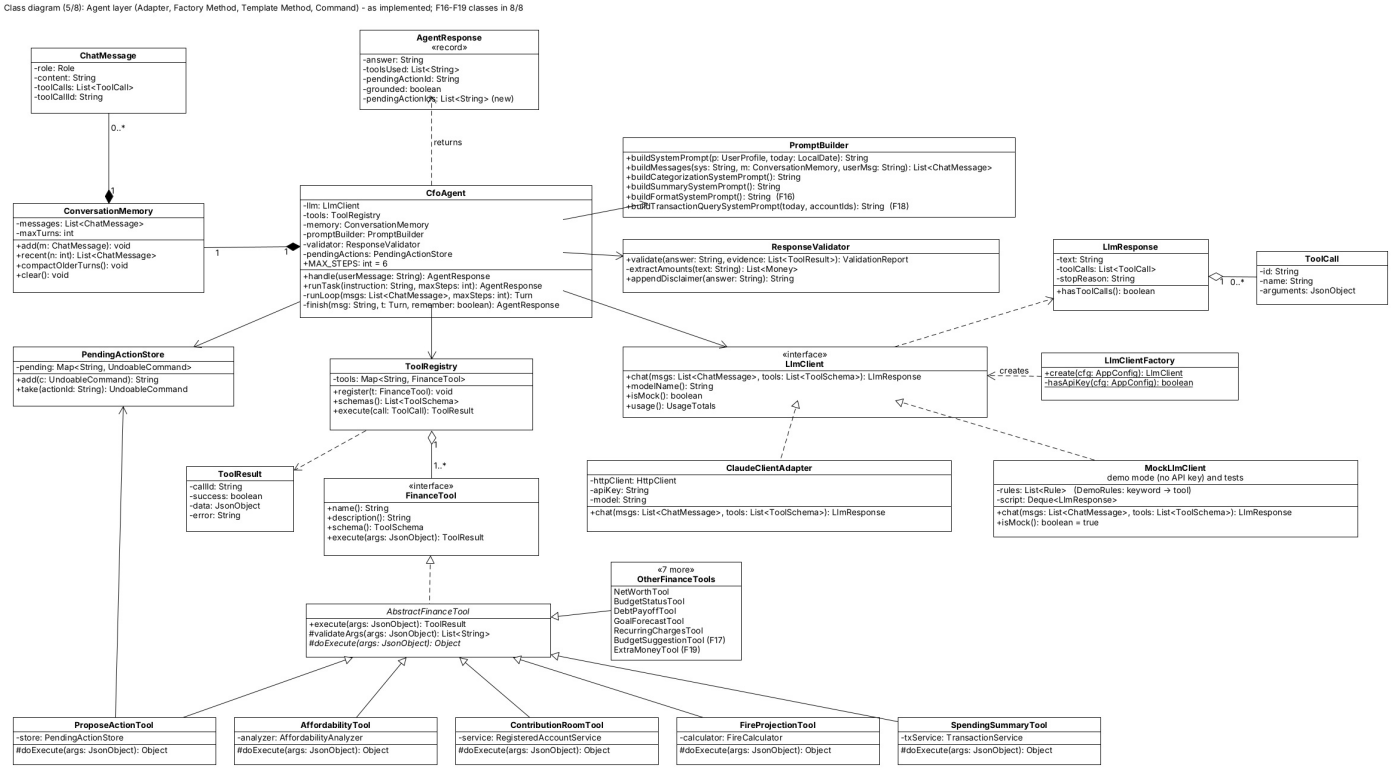


Class diagram 3/8

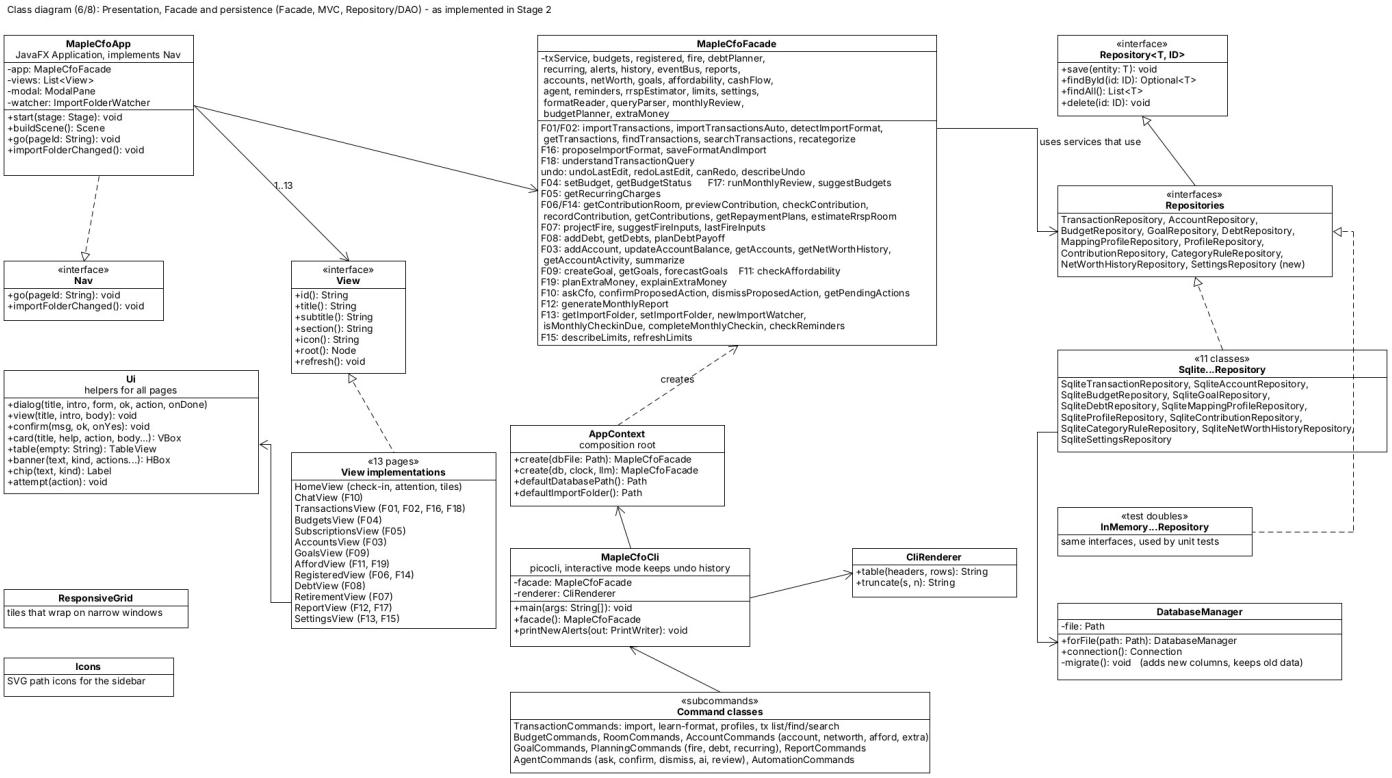
Class diagram (4/8): Events and undoable commands (Observer, Command) - as implemented in Stage 2



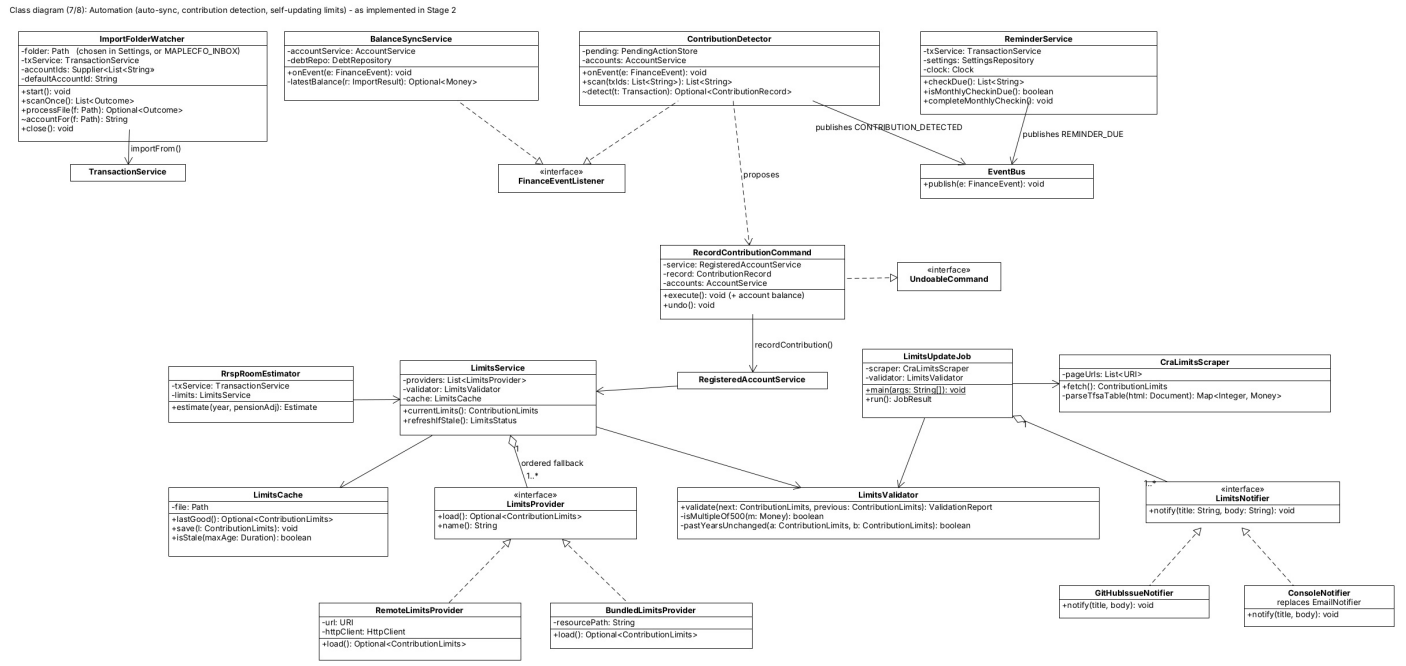
Class diagram 4/8



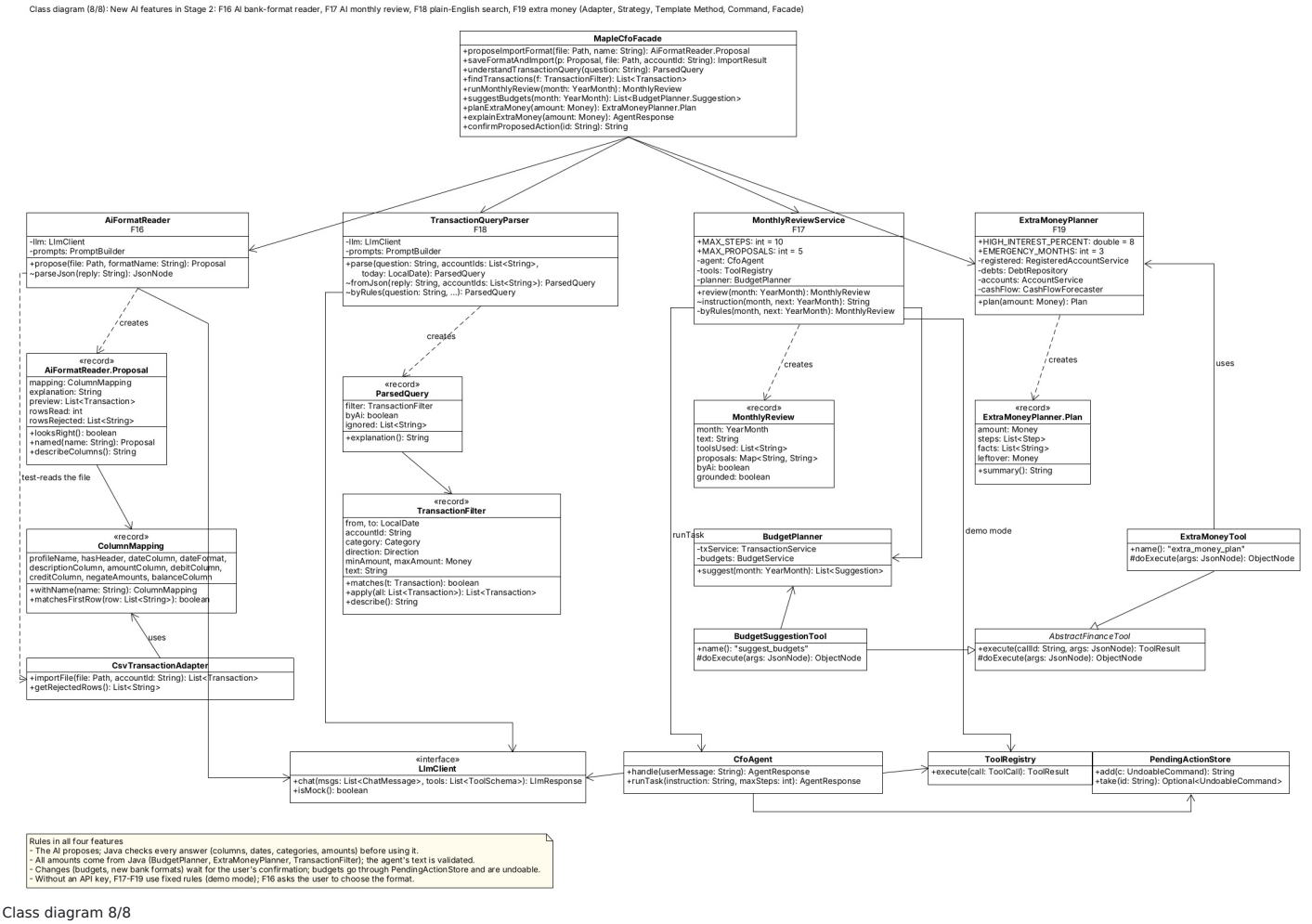
Class diagram 5/8



Class diagram 6/8

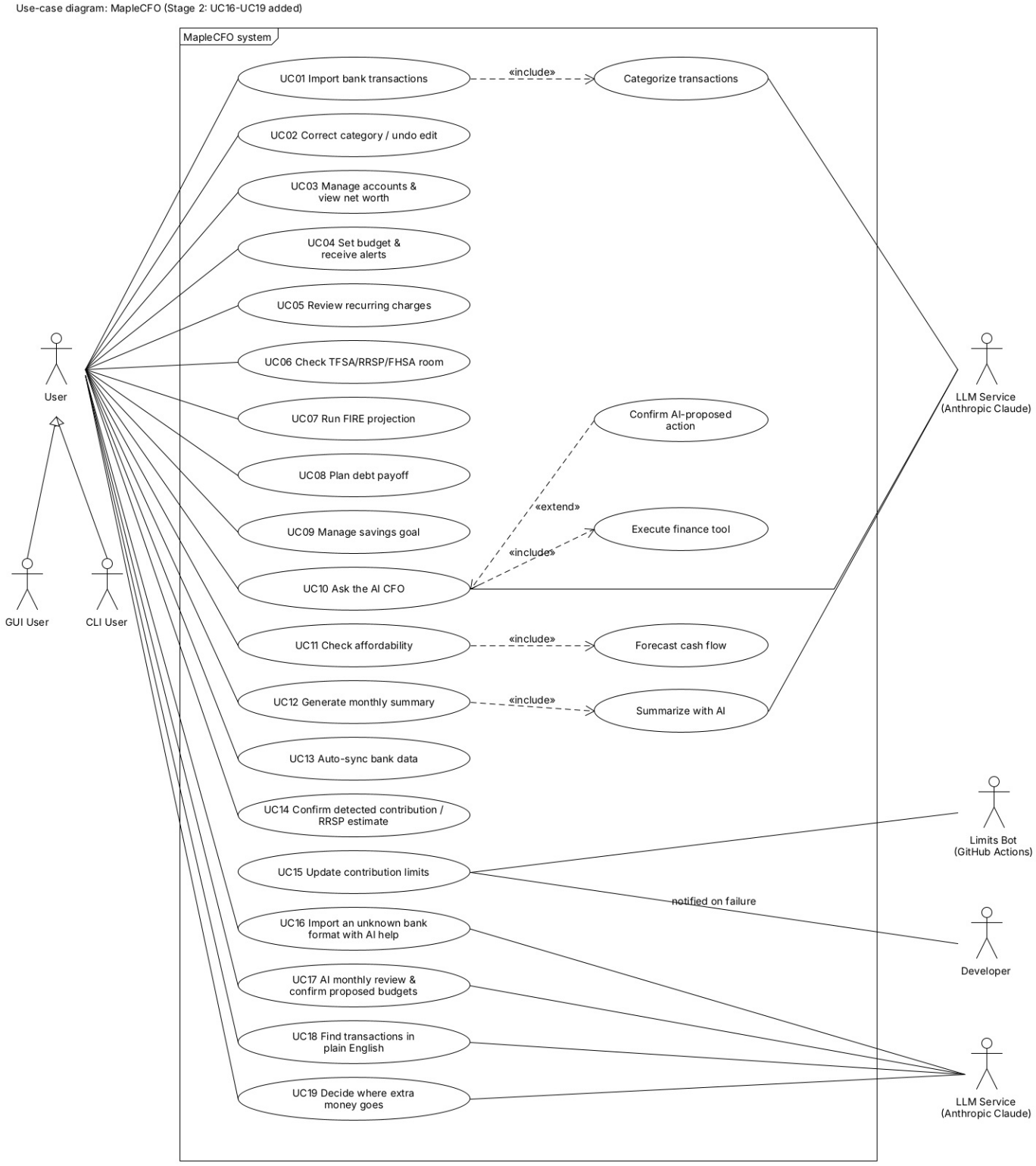


Class diagram 7/8

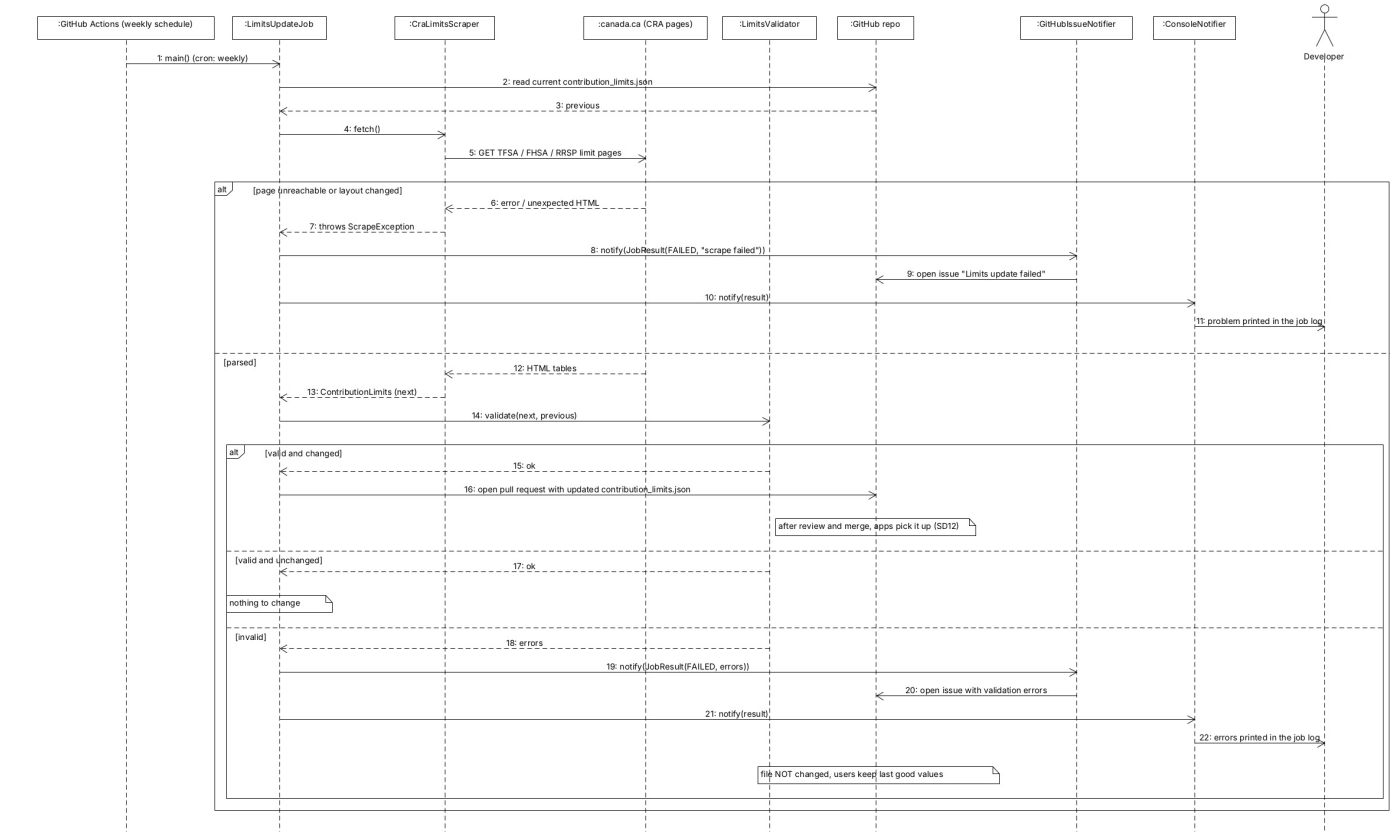


Class diagram 8/8

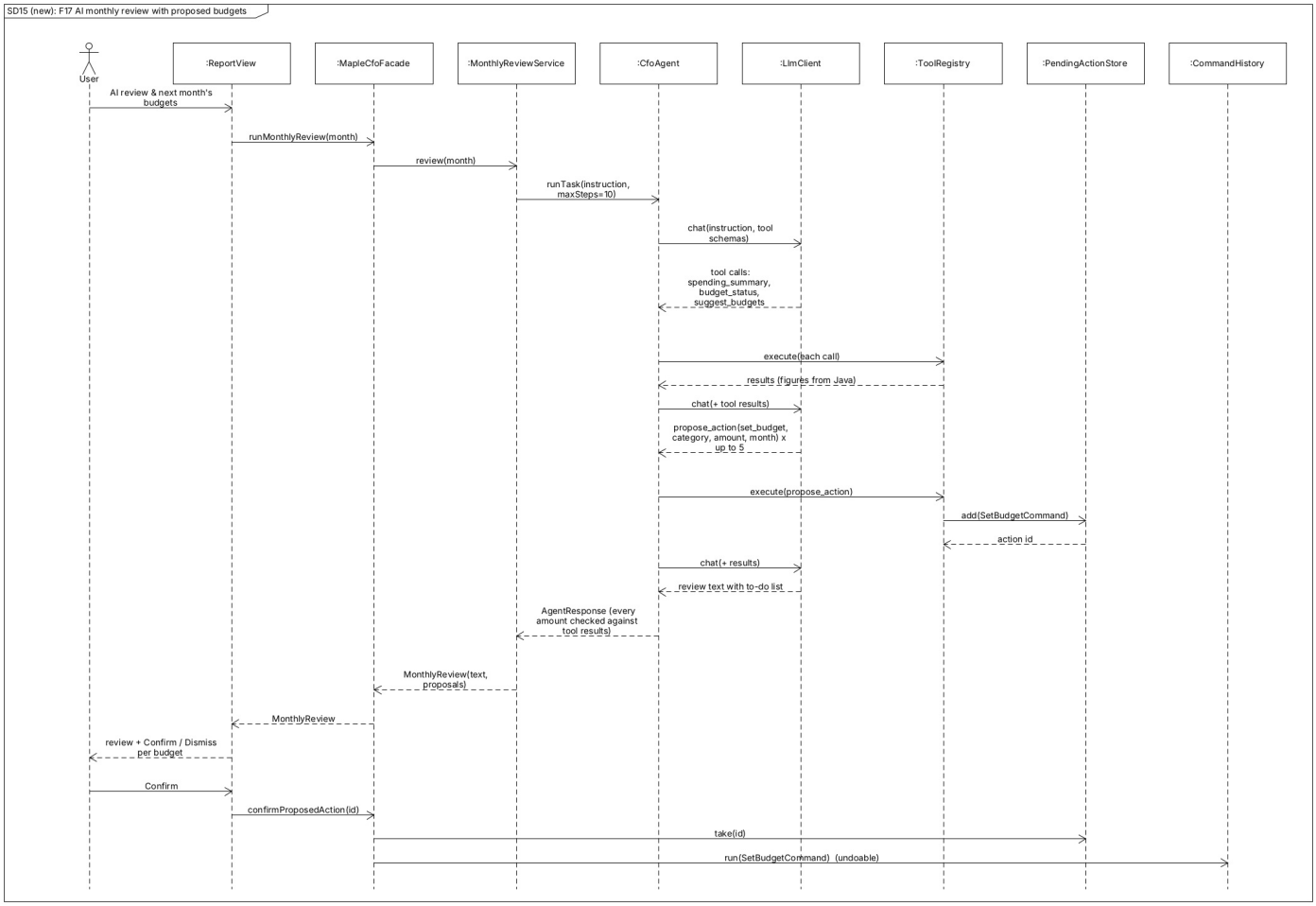
Appendix B: Use cases and new or changed sequence diagrams



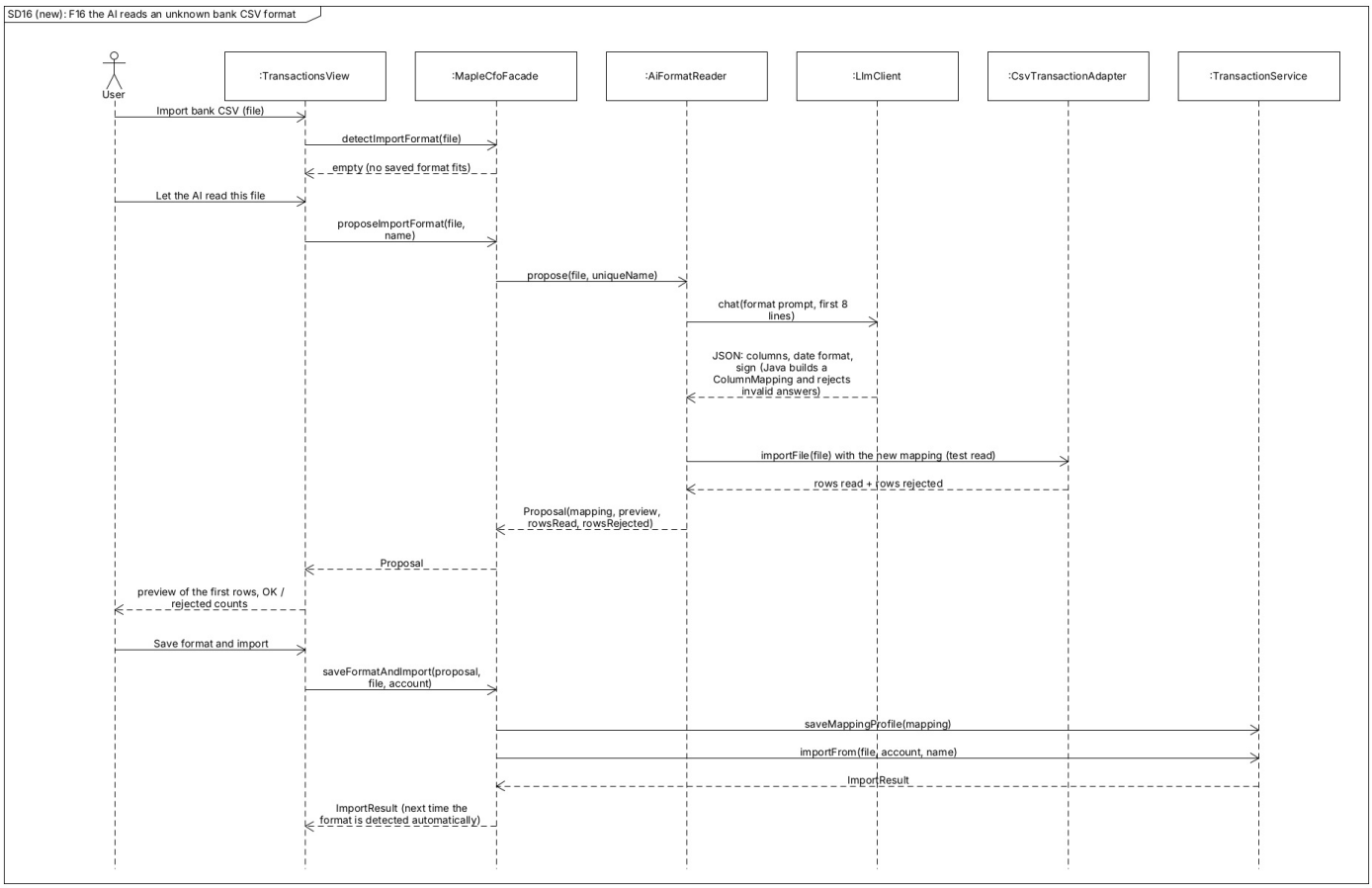
Use cases


SD13 (updated in Stage 2): Weekly limits bot - opens a pull requ
:GitHub Actions (weekly schedule)
:LimitsUpdateJob
:CraLimitsScraper
:canada.ca (CRA pages)
:LimitsValidator
:GitHub repo
:GitHubIssueNotifier
:ConsoleNotifier
Developer
1: main() (cron: weekly)
2: read current contribution_limits.json
3: previous
4: fetch()
5: GET TFSA / FHSA / RRSP limit pages
alt
[page unreachable or layout changed]
6: error / unexpected HTML
7: throws ScrapeException
8: notify(JobResult(FAILED, "scrape failed"))
9: open issue "Limits update failed"
10: notify(result)
11: problem printed in the job log
[parsed]
12: HTML tables
13: ContributionLimits (next)
14: validate(next, previous)
alt
[valid and changed]
15: ok
16: open pull request with updated contribution_limits.json
after review and merge, apps pick it up (SD12)
[valid and unchanged]
17: ok
nothing to change
[invalid]
18: errors
19: notify(JobResult(FAILED, errors))
20: open issue with validation errors
21: notify(result)
22: errors printed in the job log
file NOT changed, users keep last good values

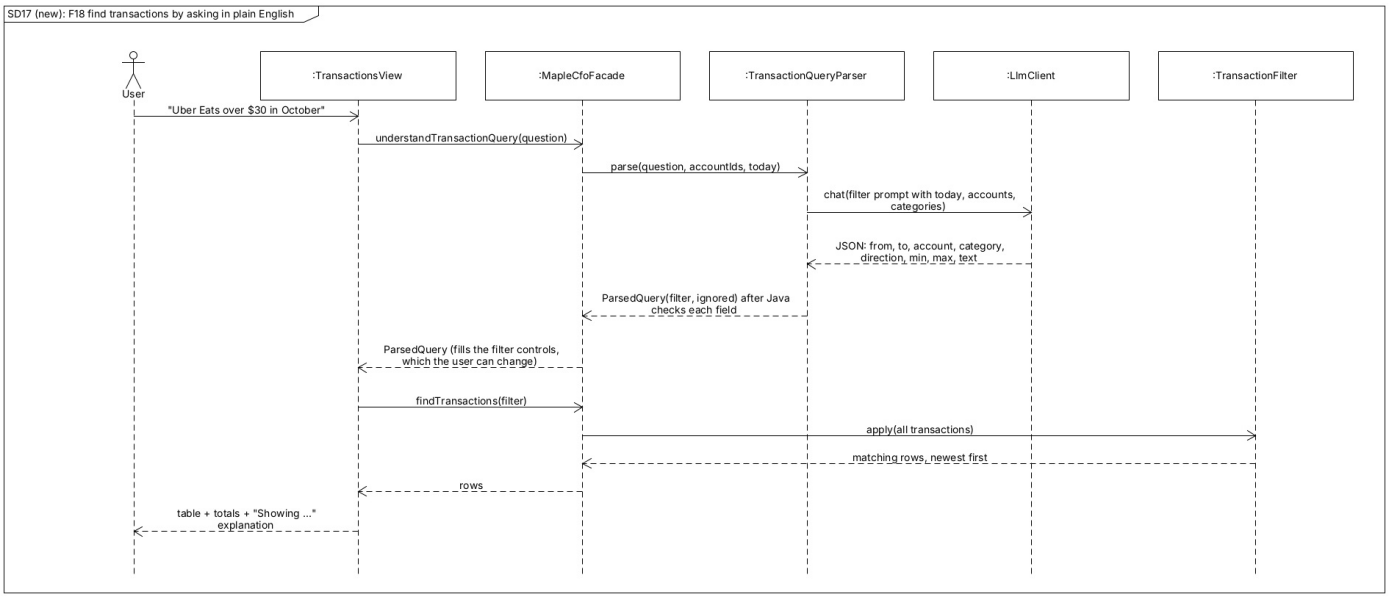
SD13


SD15 (new): F17 AI monthly review with proposed budgets
User
:ReportView
:MapleCfoFacade
:MonthlyReviewService
:CfoAgent
:LlmClient
:ToolRegistry
:PendingActionStore
:CommandHistory
AI review & next month's budgets
runMonthlyReview(month)
review(month)
runTask(instruction, maxSteps=10)
chat(instruction, tool schemas)
tool calls: spending_summary, budget_status, suggest_budgets
execute(each call)
results (figures from Java)
chat(+ tool results)
propose_action(set_budget, category, amount, month) x up to 5
execute(propose_action)
add(SetBudgetCommand)
action id
chat(+ results)
review text with to-do list
AgentResponse (every amount checked against tool results)
MonthlyReview(text, proposals)
MonthlyReview
review + Confirm / Dismiss per budget
Confirm
confirmProposedAction(id)
take(id)
run(SetBudgetCommand) (undoable)

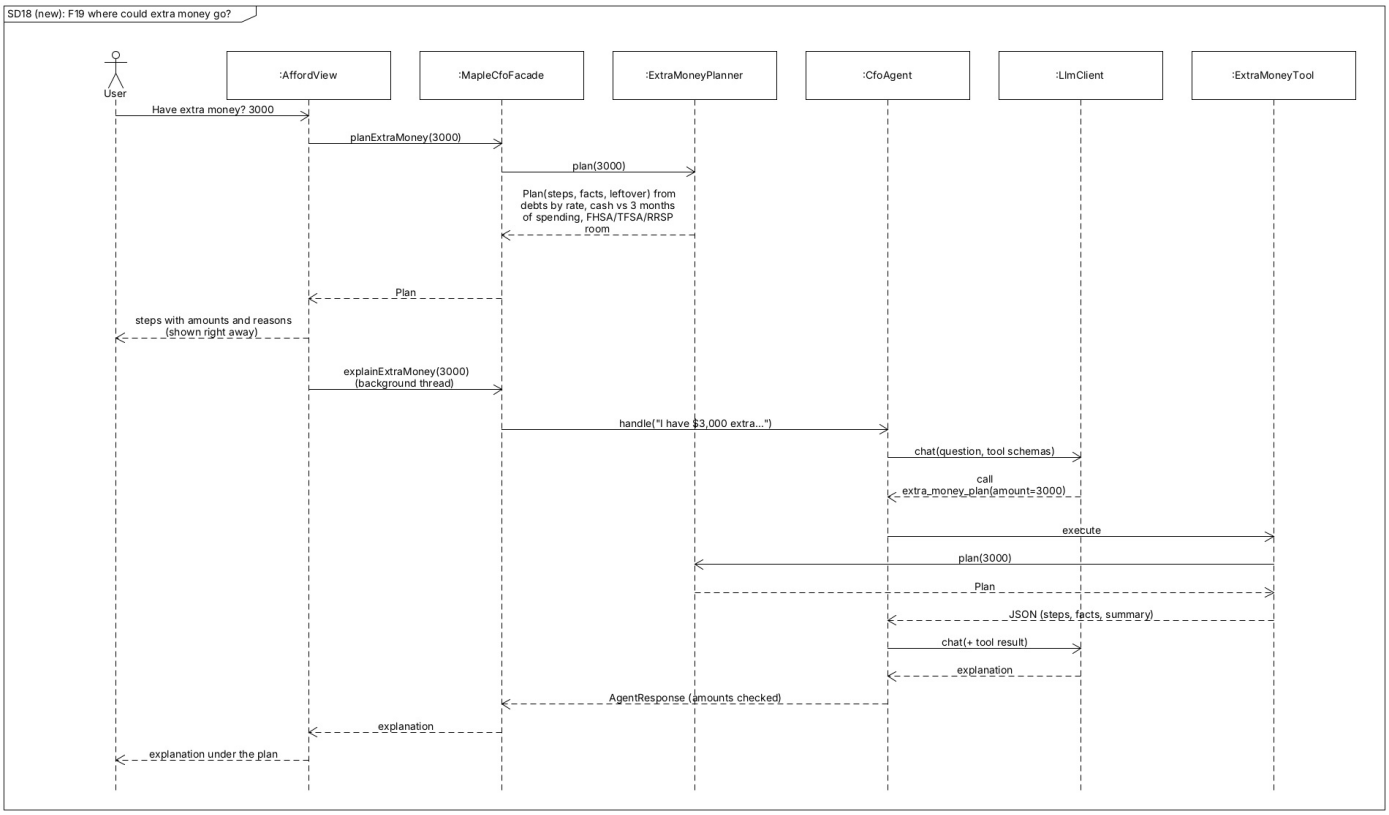
SD15


SD16 (new): F16 the AI reads an unknown bank CSV format
User
:TransactionsView
:MapleCfoFacade
:AiFormatReader
:LlmClient
:CsvTransactionAdapter
:TransactionService
Import bank CSV (file)
detectImportFormat(file)
empty (no saved format fits)
Let the AI read this file
proposeImportFormat(file, name)
propose(file, uniqueName)
chat(format prompt, first 8 lines)
JSON: columns, date format, sign (Java builds a ColumnMapping and rejects invalid answers)
importFile(file) with the new mapping (test read)
rows read + rows rejected
Proposal(mapping, preview, rowsRead, rowsRejected)
Proposal
preview of the first rows, OK / rejected counts
Save format and import
saveFormatAndImport(proposal, file, account)
saveMappingProfile(mapping)
importFrom(file, account, name)
ImportResult
ImportResult (next time the format is detected automatically)

SD16


SD17 (new): F18 find transactions by asking in plain English
User
:TransactionsView
:MapleCfoFacade
:TransactionQueryParser
:LlmClient
:TransactionFilter
"Uber Eats over $30 in October"
understandTransactionQuery(question)
parse(question, accountIds, today)
chat(filter prompt with today, accounts, categories)
JSON: from, to, account, category, direction, min, max, text
ParsedQuery(filter, ignored) after Java checks each field
ParsedQuery (fills the filter controls, which the user can change)
findTransactions(filter)
apply(all transactions)
matching rows, newest first
rows
table + totals + "Showing ..." explanation

SD17


SD18 (new): F19 where could extra money go?
User
:AffordView
:MapleCfoFacade
:ExtraMoneyPlanner
:CfoAgent
:LlmClient
:ExtraMoneyTool
Have extra money? 3000
planExtraMoney(3000)
plan(3000)
Plan(steps, facts, leftover) from debts by rate, cash vs 3 months of spending, FHSA/TFSA/RRSP room
Plan
steps with amounts and reasons (shown right away)
explainExtraMoney(3000) (background thread)
handle("I have $3,000 extra...")
chat(question, tool schemas)
call extra_money_plan(amount=3000)
execute
plan(3000)
Plan
JSON (steps, facts, summary)
chat(+ tool result)
explanation
AgentResponse (amounts checked)
explanation
explanation under the plan

SD18